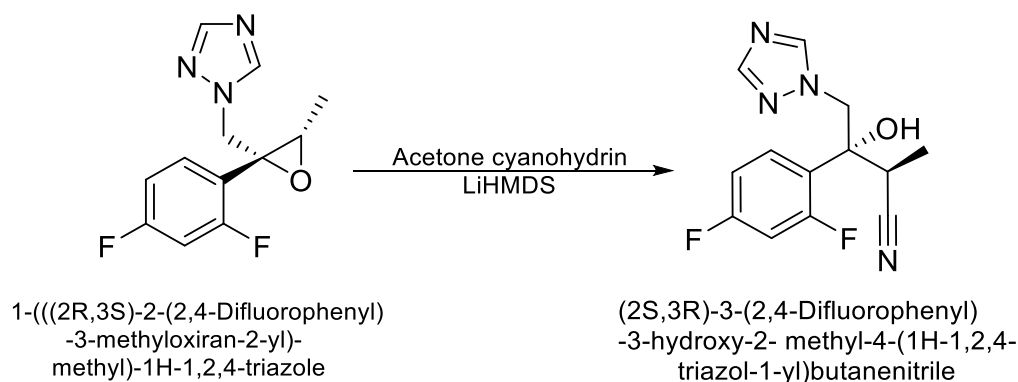


Proposed Route of Synthesis for Fosravuconazole L-Lysine Ethanolate

- 1) Organic process Research & Development 2009,vol.13,#4,p.716-728
- 2) WO2006118351 A1

Fosravuconazole Stage-01

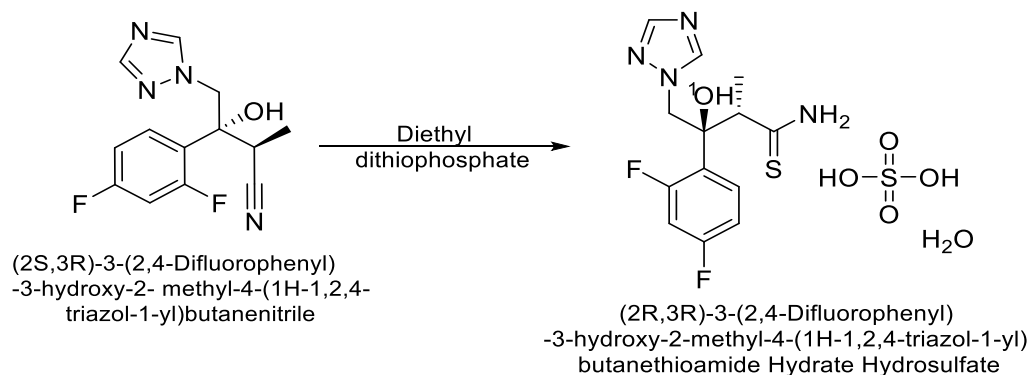


Step-1:(2S,3R)-3-(2,4-Difluorophenyl)-3-hydroxy-2-methyl-4- (1H-1,2,4-triazol-1-yl) butane nitrile

A solution of 1.65 M lithium bis(trimethylsilyl)amide in THF (190.6 kg, 352.5 mol) was distilled under vacuum at <30 °C to a volume of 140 L over 1 h and held at 0 °C for 10.5 h. Acetone cyanohydrin (37.9 kg, 441.1mol) was charged over 45 min, raising the temperature to 22°C. In another vessel, 12 (98.1 wt %, 44.2 kg, 173 mol) and THF (40 kg) were heated to 39 °C and held at that temperature until dissolution was complete. This solution was transferred to the reactor containing the mixture of lithium bis(trimethylsilyl)amide and acetone cyanohydrin over 10 min. The reactor and transfer line were washed over with THF (8 kg). The reaction mixture was heated to 65 °C and held at reflux for 12 h at which time the reaction was determined to be complete by HPLC (0.2% 12 remaining in relation to 13). After 24 h (the reaction was held an extra 12 h only to accommodate a new shift; this had been established as an acceptable hold point), the reaction was cooled to 1 °C and the pH was adjusted to 9-10 with a 13.7 wt % solution of aqueous hydrochloric acid (243 kg). During the neutralization, the temperature rose to 7°C and the product crystallized. Water (51 kg) and isopropanol (166 kg) were charged over 0.5 h. The solution was heated to 40 °C and concentrated under vacuum to 550 L. The resulting slurry was cooled to -11 °C and held for 15 h. The product was isolated in a Hastelloy centrifuge and was washed with 25%

aqueous isopropanol solution (3×15 kg) at 15°C . The cake was dried at 45°C under vacuum over 20 h to yield 43.5kg of an off-white crystalline solid (99.6 HPLC area %, 90.2%yield). An analytical sample was recrystallized from isopropanol: mp $178\text{--}182^\circ\text{C}$ (lit.4 $179\text{--}182^\circ\text{C}$). HRMS calcd M +H for $\text{C}_{13}\text{H}_{13}\text{F}_2\text{N}_4\text{O}$: 279.1052; found M + H: 279.1059. ^1H NMR (CDCl_3 , 300 MHz) 7.98 (d, J) 9.0 Hz, 1H), 7.84 (d,J) 9.0 Hz, 1H), 7.41 (m, 1H), 6.78 (m, 2H), 5.57 (s, 1H),4.97 (d, J) 14.6 Hz, 1H), 4.84 (d, J) 14.6 Hz, 1H), 3.31 (q,J) 9.0 Hz, 1H), 1.18 (d, J) 9.0 Hz, 3H); ^{13}C NMR (CDCl_3 ,126 MHz) 163.25 (dd, J) 12.7, 251.8 Hz), 157.89 (dd, J)12.7, 256.9 Hz), 152.37, 143.98, 130.97 (d, J) 5.1 Hz), 121.28(d, J) 12.7 Hz), 119.65, 112.20 (d, J) 17.8 Hz), 104.34 (t,J) 27 Hz), 75.83 (d, J) 5.1 Hz), 55.76 (d, J) 7.6 Hz),33.57, 12.86. Anal. Calcd for $\text{C}_{13}\text{H}_{12}\text{F}_2\text{N}_4\text{O}$: C, 56.11; H, 4.35;F, 13.66; N, 20.13. Found: C, 56.10; H, 4.23; F, 13.35; N, 20.09.

Fosravuconazole Stage-02

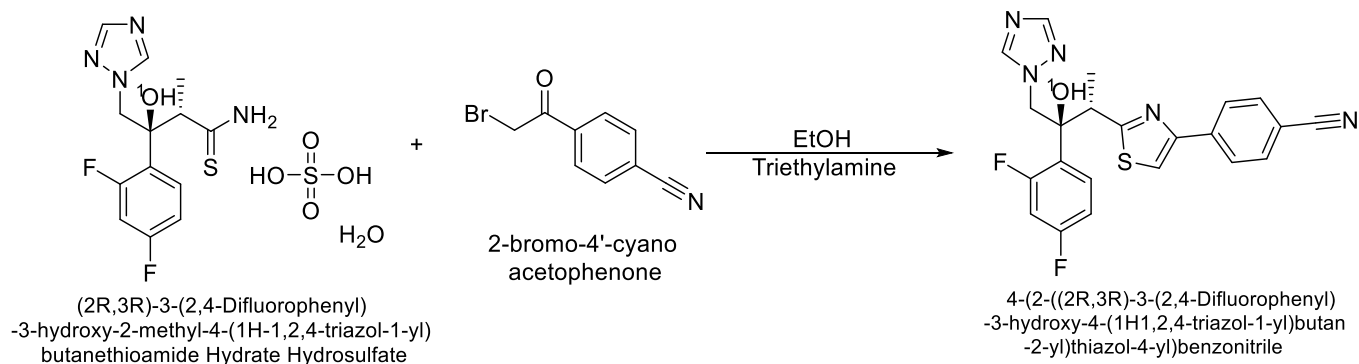


Step-2:(2R,3R)-3-(2,4-Difluorophenyl)-3-hydroxy-2-methyl-4-(1H-1,2,4-triazol-1-yl)butanethioamide Hydrate Hydrosulfate

A solution of 13 (43.0 kg, 154.5 mol), water (17.2 L), and isopropyl alcohol (54.0 kg) was heated to 78°C over 80 min at which point a solution had occurred. Diethyl dithiophosphate (86.2 kg, 462.9 mol) was added over 1.5 h while maintaining the temperature at $75\text{--}80^\circ\text{C}$. (CAUTION: diethyl dithio phosphate is highly toxic, corrosive, and combustible and produces a stench. This reaction also leads to off-gasing of hydrogen sulfide, which is highly flammable, toxic, and irritating and produces a stench. Proper precautions are required prior to conducting this reaction.) Additional isopropanol (2 kg) was used to rinse the line into the reactor. The heating was maintained for 7.5 h, until reaction completion (13) 0.93 HPLC area %). The mixture was cooled

to 20 °C over 1 h. Ethyl acetate (194 kg) and water (215 kg) were charged to the mixture over 50 min, and the mixture cooled to 2 °C over 80 min. Sodium hydroxide solution (3N, 284 kg) was charged over 75 min while maintaining the resulting temperature at 10-20 °C. The final pH was 8.1, and the final temperature 15 °C. The phases were separated, and the organic phase was treated with 20 wt % sodium hydroxide (259 kg). The mixture was stirred for 30 min at ambient temperature (23 °C), and the layers were separated. The transfer lines were rinsed with ethyl acetate (2 kg). The combined organic phases were then treated with 98 wt % sulfuric acid (15.3 kg, 152.9 mol) over 3 min at a final temperature of 40 °C. The lines were rinsed with ethyl acetate (2 kg). The solution was concentrated to 87 L by vacuum distillation at 50-60 °C over 6 h. The slurry was cooled to 26 °C over 70 min, and MTBE (128 kg) was charged over 9 min. The slurry was further cooled to 5 °C over 1.5 h and held at 0.2-5 °C over 1 h. The crystals were collected by filtration and washed with MTBE (136 kg). The salt was dried at 60 °C under vacuum for 34 h to achieve an LOD of 0.35%. A total of 58.2 kg were isolated in 91.2% yield (99.9 HPLC area %). An analytical sample was recrystallized from ethyl acetate/ MTBE. Mp) 152 °C (dec.). Anal. Calcd for C₁₃H₁₈F₂N₄O₆S₂: C, 36.44; H, 4.23; N, 13.08; F, 8.76; S, 14.97. Found: C, 36.41; H, 4.07; N, 13.13; F, 9.14; S, 15.02. 1 H NMR (DMSO-d₆, 300 MHz) (not all exchangeable protons detected) 10.19 (d, J) 46.1 Hz, 2H), 8.85 (s, 1H), 7.98 (s, 1H), 7.28 (m, 2H), 6.92 (m, 1H), 6.70 (br s, 1H), 4.84 (d, J) 14.1 Hz, 1H), 4.57 (d, J) 14.1 Hz, 1H), 3.60 (q, J) 6.6 Hz, 1H), 0.92 (d, J) 6.8 Hz, 3H). 13C NMR (DMSO-d₆, 76 MHz) 208.3, 162.7 (d, J) 250 Hz), 157.1 (d, J) 264 Hz), 146.7, 143.9, 130.7, 123.3, 111.6, 104.2, 75.2, 57.3, 49.6, 16.3.

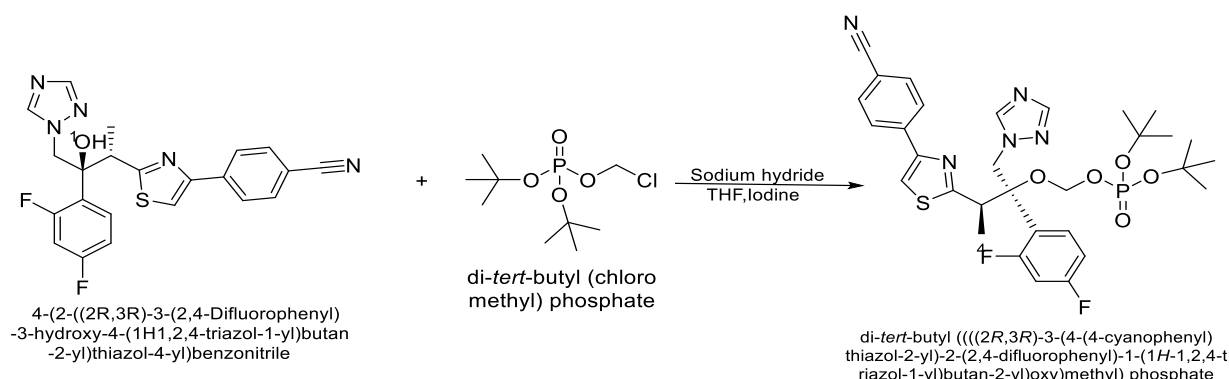
Fosravuconazole Stage-03



Step-3: 4-(2-((2R,3R)-3-(2,4-Difluorophenyl)-3-hydroxy-4-(1H-1,2,4-triazol-1-yl)butan-2-yl)thiazol-4-yl)benzonitrile

A mixture of 15 (25.8 kg, 99.4 area % purity, 60.2 mol adjusted for water content), 2-bromo-4'-cyanoacetophenone (hazard, this is a powerful lachrymator and requires PPE as described for the previous step) (13.8 kg, 61.9 mol), and 95% ethanol (104 kg) was heated to 60-70 °C and held for 2 h until the reaction was complete (0.09 HPLC area % of the aryl bromide relative to 15). Water (122 kg) and 95% ethanol (104 kg) were charged at 55-70 °C. The pH of the solution was adjusted with triethylamine (15.6 kg, 154.2 mol) to pH 3.5-4.0. The solution was stirred at 55-65 °C for 20 min followed by cooling to 20 °C over 2 h. The slurry was stirred for 10 h at 20-25 °C, and the crystals were isolated in a Hastelloy filter dryer. The cake was washed twice with a 50 v/v% ethanol/water solution (98 kg ethanol, 123 kg water) and vacuum-dried at 50-55 °C for 8 h to yield 22.7 kg of a white, crystalline solid (99.7 HPLC area %, LOD 0.12%, 85.8% yield).

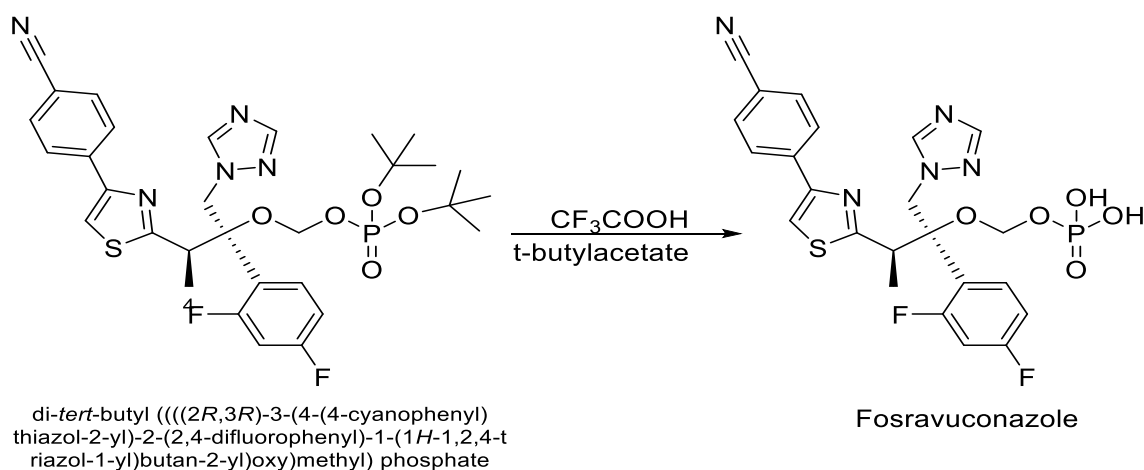
Fosravuconazole Stage-04



An oven dried, 1L round-bottom flask equipped with a mechanical stirrer, nitrogen inlet adapter, pressure-equalizing addition funnel fitted with a rubber septum and temperature probe was charged with sodium hydride (2.89 g, 0.069 mol, 60%) and THF (50 mL). To this stirred suspension, (2R,3R)-3-[4-(4-cyanophenyl)thiazol-2-yl]-2-(2,4-difluorophenyl)-1-(1H-1,2,4-triazol-1-yl)butan-2-ol, II, (10 g, 0.023 mol) in 30 mL of THF was added dropwise over 20 minutes at room temperature. After stirring for 45 minutes, a solution of iodine (2.99 g, 0.0115 mol) in THF (30 mL) was added dropwise over 10 minutes followed by dropwise addition of compound di tert butylchloromethyl phosphate, III (13.29 g, 0.035 mol, ~68% purity) over 15 minutes. The reaction

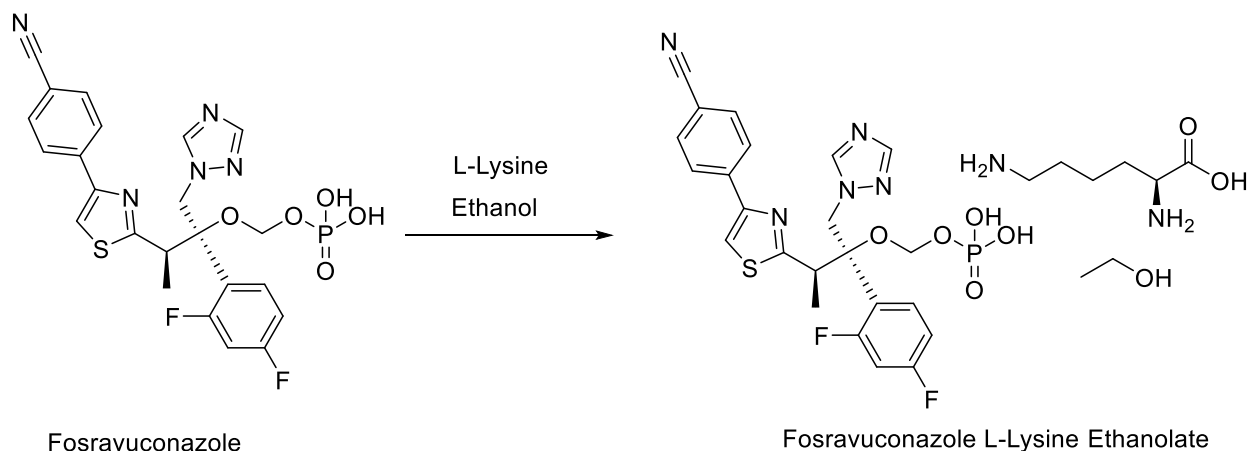
mixture was stirred for 4 hours at about 41° C. to complete the reaction. The completion of the reaction was judged by in-process HPLC. The reaction mixture was poured into ice cold water (100 mL). The aqueous phase was separated and extracted with ethyl acetate (3×50 mL) and the combined organic extract was washed with 10% sodium thiosulfite (50 mL), water (50 mL), brine (50 mL), dried over magnesium sulfate and concentrated under reduced pressure to give pale yellow oil (22.8 g, In-process HPLC: ~97% pure).

Fosravuconazole Stage-05



To a round-bottom flask equipped with magnetic stirrer, cooling bath, pH probe and N₂ inlet-outlet was charged the product of Step A above (7.5 g) in CH₂Cl₂ (23 mL) and cooled to 0° C. To this stirred solution, trifluoroacetic acid (8.8 mL) was added slowly and stirred for 3 h to complete the reaction. The completion of the reaction was judged by in-process HPLC. The reaction mixture was poured into a cold solution of 2N NaOH (64 mL). The reaction mixture was extracted with *t*-butyl acetate (2×65 mL) to remove all the organic impurities. The aqueous layer containing the title product as bis sodium salt was treated with activated charcoal (10 g) and filtered through a bed of Celite. The clear filtrate was acidified with 1N HCl to pH 2.5. The free acid, the title product, was extracted into ethyl acetate (2×50 mL). The combined organic layer was washed with water, dried over MgSO₄, filtered, and the filtrate concentrated under reduced pressure to afford 3.39 g of crude title product.

Fosravuconazole Stage-06



The above obtained product V was dissolved in methanol (75 mL) . With the free acid solution, L-lysine (1.8 g) was added with the pH maintained at 4.2 to 5.5, and the mixture was heated at 60°C for 4.5 h. The hot reaction mixture was filtered through a bed of Celite. The filtrate was concentrated to about 5 πiL, mixed with ethanol (100 mL) and heated to 65°C to crystallize the solvate of the mono-lysine salt. The solvate was collected on a Buchner funnel and dried under vacuum to afford 3.71 g of the title solvate compound as a crystalline solid.